Platyspondylic Dysplasia, Torrance Type (PLSD-T) — Comprehensive Disease Report

Disease: Platyspondylic Dysplasia, Torrance type (PLSD-T) **MONDO:** MONDO:0007895 · **OMIM:** 151210 · **Causal gene: COL2A1** (type II collagen, alpha-1 chain; HGNC:2200; chromosome 12q13.11) **Category:** Mendelian, autosomal dominant

Summary

Platyspondylic Dysplasia, Torrance type (PLSD-T) is a rare, usually perinatally lethal skeletal dysplasia caused by heterozygous **dominant-negative mutations in the C-propeptide (COOH-/NC2 propeptide) domain of the *COL2A1* gene**, which encodes the alpha-1 chain of type II (cartilage) collagen. It is therefore a **type II collagenopathy**, and specifically defines a distinct **C-propeptide subfamily** within that group of disorders, separate from the more numerous triple-helical-domain type II collagenopathies (achondrogenesis type II, hypochondrogenesis, spondyloepiphyseal dysplasia congenita, Kniest dysplasia, Stickler syndrome). The cardinal features are **platyspondyly** (flat, severely under-ossified vertebral bodies), **micromelia** (extremely short limbs), **brachydactyly**, and characteristic **metaphyseal changes**, with a small thorax causing lethal pulmonary hypoplasia. Most affected infants are stillborn or die in the neonatal period, though rare long-term survivors have been reported.

The mechanistic basis is well established and unusually clear for a rare dysplasia. Unlike loss-of-function/haploinsufficiency mechanisms, the C-propeptide mutations (missense, last-exon nonsense variants that escape nonsense-mediated decay, and frameshift variants) produce an **altered pro-alpha-1(II) chain** that cannot assemble/trimerize and secrete normally. The mutant chains are **retained in the chondrocyte endoplasmic reticulum (ER)**, triggering **ER stress and the unfolded protein response (UPR)** (upregulation of *Grp94*/HSP90B1 and *Chop*/DDIT3), **expanded/distended rough ER**, and **growth-plate chondrocyte apoptosis**. The combination of defective cartilage matrix and chondrocyte death disrupts endochondral

ossification, producing the skeletal phenotype. This causal chain has been directly demonstrated in three independent semidominant *Col2a1* C-propeptide mouse models (p.D1469A, p.Y1391S, and the classic disproportionate micromelia *Dmm* allele).

PLSD-T lies on an **allelic phenotypic continuum** with the milder, viable **spondyloperipheral dysplasia (SPD)**: identical C-propeptide mutations (e.g., Y1391C) have been reported in both lethal PLSD-T and surviving SPD patients. Familial recurrence from phenotypically normal parents is explained by **germline (germinal) mosaicism**, and a documented mother–fetus pair sharing the same mutation illustrates **somatic mosaicism** producing mild SPD in the mosaic parent and lethal PLSD-T in the fully-affected offspring. There is no disease-modifying therapy; management is supportive/palliative, and the mainstays of care are accurate diagnosis, genetic counseling, and reproductive risk management.

Key Findings

Finding 1 — PLSD-T is caused by dominant-negative C-propeptide mutations in *COL2A1*

The defining molecular study (Zankl et al., 2005) examined eight additional PLSD-T cases (plus two previously reported) and found that **all** had mutations localized to the **C-propeptide domain** of *COL2A1*. The mutational spectrum includes **missense, nonsense (stop-codon), and frameshift** mutations. Critically, all nonsense mutations fell within the **last exon**, where they escape nonsense-mediated mRNA decay (NMD) — meaning a stable, altered protein is produced rather than a null allele. This establishes PLSD-T as a **dominant-negative** disorder of collagen chain assembly, not haploinsufficiency, and defines it as a "**novel subfamily within the type 2 collagenopathies**."

> *"We studied eight additional cases of PLSD-T and found that all had mutations in the C-propeptide domain of COL2A1. The mutational spectrum includes missense, stop codon and frameshift mutations. All non-sense mutations were located in the last exon, where they would escape non-sense-mediated RNA-decay."* — PMID: 15643621

Mechanistic significance: The C-propeptide is essential for correct selection and nucleation of the three pro-alpha chains into the procollagen trimer. A mutation here produces a chain that incorporates into (or blocks) trimer assembly, poisoning the product of the normal allele — the hallmark of a dominant-negative effect.

Finding 2 — ER stress-mediated chondrocyte apoptosis is the mechanistic driver

Two independent *Col2a1* ENU-induced mutant mouse lines carrying C-propeptide mutations (**p.D1469A** and **p.Y1391S**) at residues corresponding to human PLSD-T mutations recapitulated the lethal skeletal dysplasia when homozygous (extremely short limbs, severe spondylar and pelvic dysplasia, brachydactyly). In these mice, secretion of the mutant collagen into the extracellular space was disrupted, the rough ER was abnormally expanded, ER-stress/UPR genes (*Grp94*, *Chop*) were upregulated, and **chondrocyte apoptosis was severely induced** in the growth plate.

> *"The secretion of the mutant proteins into the extracellular space was disrupted, accompanied by an abnormally expanded endoplasmic reticulum (ER) and the up-regulation of ER stress-related genes in chondrocytes. Chondrocyte apoptosis was severely induced in the growth plate of the homozygotes."* — PMID: 26545783
>
> *"The secretion of the mutant proteins into the extracellular space was disrupted, accompanied by abnormally expanded rough endoplasmic reticulum (ER) and upregulation of ER stress-related genes, such as Grp94 and Chop, in chondrocytes."* — PMID: 21538020

Mechanistic significance: This provides direct in-vivo evidence that **ER retention → UPR activation → chondrocyte apoptosis** is the pathway linking the mutation to growth-plate failure, in two independently-derived models.

Finding 3 — PLSD-T and spondyloperipheral dysplasia form an allelic spectrum with germline/somatic mosaicism

The same C-propeptide mutations cause lethal PLSD-T and the milder, viable **spondyloperipheral dysplasia (SPD)**; for example, the **Y1391C** substitution has been reported in both conditions, directly demonstrating a phenotypic continuum. Recurrence in two siblings born to phenotypically normal parents (c.3545G>A, p.Gly1182Asp) proved **germinal**

(germline) mosaicism in a parent. A separate mother–fetus pair sharing c.4458_4460delCTT (p.Phe1486del) demonstrated **somatic mosaicism**: the mosaic mother had mild SPD while her non-mosaic fetus had lethal PLSD-T.

> *"the presence of the same mutation in the two sibs is proof of germinal mosaicism in one of the parents"* — PMID: 22711552
>
> *"The phenotype in the mother was compatible with mild spondyloperipheral dysplasia (SPD). Molecular studies documented somatic mosaicism for the same mutation in the mother."* — PMID: 22495950

Clinical significance: This explains apparent "autosomal recessive"-looking recurrence in sporadic dominant disease, and has direct recurrence-risk and genetic-counseling implications. The dosage/mosaicism dependence of severity is a recurring theme across the human and mouse data.

Finding 4 — Multiple *Col2a1* C-propeptide mouse models recapitulate the disease via impaired procollagen assembly/secretion

Three semidominant mouse lines carry *Col2a1* C-propeptide mutations: the ENU alleles **p.D1469A** (P 21538020) and **p.Y1391S** (P 26545783), and the classic **disproportionate micromelia (*Dmm*)** allele (P 18678883). Homozygotes show lethal dwarfism with short limbs and spondylar dysplasia; heterozygotes show mild dwarfism — a clean **gene-dosage-dependent severity gradient**. Ultrastructurally, chondrocytes show distended rough ER, impaired assembly/secretion of type II procollagen trimers, thinner and sparser cartilage fibrils, and reduced aggrecan mRNA.

> *"These findings support the prevailing hypothesis that a C-propeptide mutation could interrupt the normal assembly and secretion of Type II procollagen trimers"* — PMID: 18678883

> *"The phenotype was inherited as a semidominant trait. The heterozygotes were mildly but significantly smaller than wild-type mice. The homozygotes exhibited lethal skeletal dysplasias"* — PMID: 21538020

Significance: These faithful models confirm the human mechanism and provide a platform for testing interventions targeting ER stress/UPR.

Finding 5 — PLSD-T clinical/radiographic phenotype: platyspondyly, micromelia, brachydactyly, metaphyseal changes, perinatal lethality

PLSD-T is defined clinically and radiographically by **platyspondyly, brachydactyly, and metaphyseal changes**, and is generally **perinatally lethal**, though a few long-term survivors are on record. Fetal descriptions emphasize **platyspondyly, extremely short limbs, and mild brachydactyly**. The distinctive radiology — metaphyseal changes plus brachydactyly — is what separates this C-propeptide subfamily from the helical-domain type II collagenopathies.

> *"Platyspondylic lethal skeletal dysplasia (PLSD) Torrance type (PLSD-T) is a rare skeletal dysplasia characterized by platyspondyly, brachydactyly, and metaphyseal changes. Generally a perinatally lethal disease, a few long-term survivors have been reported."* — PMID: 15643621
>
> *"a rare skeletal dysplasia characterized by platyspondyly, extremely short limbs, and mild brachydactyly"* — PMID: 22495950

Report by Section

1. Disease Information

PLSD-T is a rare, usually lethal **chondrodysplasia / skeletal dysplasia** of the type II collagenopathy group. It is characterized by severe platyspondyly, extremely short limbs (micromelia), brachydactyly, and characteristic metaphyseal changes, with a narrow thorax leading to pulmonary hypoplasia and perinatal death.

Key identifiers: - **OMIM:** 151210 - **MONDO:** MONDO:0007895 - **Gene:** *COL2A1* (OMIM 120140; HGNC:2200; NCBI Gene 1280; UniProt P02458) - **Orphanet:** Platyspondylic dysplasia, Torrance type (within the type II collagenopathy / lethal platyspondylic dysplasia group) - **ICD-10:** Q77.x (chondrodysplasia with defects of growth of tubular bones and spine); **ICD-11:** LD24 (chondrodysplasias) - **MeSH:** closest terms — Osteochondrodysplasias; Dwarfism

Synonyms: platyspondylic skeletal dysplasia, Torrance type; PLSD-T; Platyspondylic dysplasia, Torrance-Luton type; Platyspondylic lethal skeletal dysplasia, Torrance type.

Information source: Aggregated disease-level resources (OMIM, Orphanet) combined with primary case reports and small case series (individual patients/fetuses). There is no large EHR-based cohort given the rarity and lethality.

2. Etiology

- **Primary cause:** Genetic — heterozygous **dominant-negative mutations in the C-propeptide domain of *COL2A1*** (Finding 1). The condition is monogenic and Mendelian; no environmental or infectious etiology.
- **Genetic risk factors:** The causal variants themselves are the risk factor. Most cases arise from **de novo** mutations; **germline and somatic mosaicism** in a parent can cause recurrence or mild parental phenotypes (Finding 3).
- **Environmental risk factors:** None identified. As with most de novo dominant disorders, advanced parental age is a theoretical contributor to de novo mutation rate but is not specifically established for PLSD-T.
- **Protective factors:** None known (genetic or environmental). Not applicable for a fully-penetrant dominant-negative lethal disorder.
- **Gene–environment interactions:** None established; the phenotype is essentially determined by the specific mutation and its dosage/mosaic status.

3. Phenotypes

All phenotypes are **congenital/neonatal**, **severe**, and generally **non-progressive** (the disorder is lethal in the perinatal period, so progression is largely moot). Frequencies are "typical/near-universal" in affected individuals given the small reported case numbers.

Phenotype	Type	HPO term (suggested)	Onset / severity / frequency
Platyspondyly (flat vertebral bodies)	Radiographic sign	HP:0000926	Congenital; severe; near-universal
Micromelia / extremely short limbs	Physical manifestation	HP:0002983 (micromelia) / HP:0003097 (short femur)	Congenital; severe; near-universal
Brachydactyly	Physical manifestation	HP:0001156	Congenital; mild–moderate; frequent
Metaphyseal changes/dysplasia	Radiographic sign	HP:0000944	Congenital; characteristic; frequent
Narrow/small thorax	Clinical sign	HP:0000774 (narrow chest)	Congenital; severe
Pulmonary hypoplasia	Clinical/ pathologic	HP:0002089	Congenital; cause of lethality
Perinatal/neonatal lethality	Outcome	HP:0003811 (neonatal death)	Usual outcome
Short stature / dwarfism (survivors)	Physical manifestation	HP:0004322	In rare survivors; severe disproportionate
Macrocephaly (reported in SPD end)	Physical manifestation	HP:0000256	Variable

Quality-of-life impact: For the lethal presentation, QoL metrics are not applicable (perinatal death). For rare survivors and for the milder SPD end of the spectrum, expect profound disproportionate short stature, orthopedic/spinal complications, and respiratory compromise with substantial impact on mobility and daily function. No disease-specific EQ-5D/SF-36 data exist.

4. Genetic/Molecular Information

- **Causal gene:** *COL2A1* (12q13.11), encoding pro-alpha-1(II) collagen chain.
- **Pathogenic variants:** Located in the **C-propeptide (COOH-/NC2) domain**, encoded largely by the final exons. Variant classes: **missense**, **nonsense** (last-exon, NMD-escaping), and

frameshift. Representative reported variants: **Y1391C/Y1391S**, **D1469A** (mouse-equivalent residue), **p.Gly1182Asp (c.3545G>A)**, **p.Phe1486del (c.4458_4460delCTT)** (Findings 1, 3). Per ACMG/AMP, such variants are classified **pathogenic/likely pathogenic** given gene–phenotype match, functional data, and de novo occurrence.

- **Allele frequency:** Absent from population databases (gnomAD); these are private, disease-causing variants, consistent with a lethal dominant disorder.
- **Somatic vs germline: Germline** (constitutional) in affected offspring; parental **germline or somatic mosaicism** documented (Finding 3).
- **Functional consequence: Dominant-negative** (misassembly + ER retention), NOT loss of function/haploinsufficiency.
- **Modifier genes:** None established. Severity appears driven chiefly by the specific mutation and mosaic dosage rather than trans-acting modifiers.
- **Epigenetics:** No disease-specific methylation/histone data reported.
- **Chromosomal abnormalities:** None; this is a single-nucleotide/small-indel disorder, not a structural/aneuploidy syndrome.

5. Environmental Information

No environmental, lifestyle, or infectious contributing factors are known or expected. PLSD-T is a purely genetic, de-novo-dominant disorder (or inherited via parental mosaicism). Not applicable for toxins, radiation, diet, or infectious agents.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **heterozygous C-propeptide mutation in *COL2A1*** (missense, last-exon nonsense, or frameshift) **produces an altered, stable pro-alpha-1(II) chain** (not a null allele; nonsense variants escape NMD). *[demonstrated —*

   15643621

   ]
2. The altered C-propeptide **impairs correct chain selection, trimerization and assembly of the type II procollagen trimer** (dominant-negative poisoning of normal chains). *[demonstrated in Dmm mouse —*

   18678883

   ]

3. Misassembled procollagen **cannot be secreted and is retained in the chondrocyte rough endoplasmic reticulum**, which becomes **abnormally distended/expanded**. *[demonstrated —*

   P *21538020*

   P *26545783* *]*

4. ER accumulation of misfolded protein **triggers the unfolded protein response / ER stress**, with **upregulation of Grp94 (HSP90B1) and Chop (DDIT3)**. *[demonstrated —*

   P *21538020*

   P *26545783* *]*

5. Sustained ER stress **leads to growth-plate chondrocyte apoptosis**. *[demonstrated —*

   P *26545783* *]*

6. Chondrocyte death **plus a deficient/abnormal cartilage extracellular matrix** (thinner, sparser type II collagen fibrils; reduced aggrecan) **disrupts the cartilage growth plate and endochondral ossification**. *[demonstrated —*

   P *18678883* *]*

7. Disrupted endochondral ossification **results in platyspondyly, micromelia, brachydactyly and metaphyseal dysplasia**, and a **small thorax**. *[demonstrated —*

   P *15643621* *]*

8. The small thorax **causes pulmonary hypoplasia and respiratory failure → perinatal lethality**. *[inferred from phenotype; standard for lethal platyspondylic dysplasias]*

Branch: The severity of the chain depends on **mutant dosage** — full constitutional heterozygosity → lethal PLSD-T; **mosaicism** (lower effective mutant load) → milder, viable SPD. In mice, heterozygotes are mildly affected while homozygotes are lethal (semidominant), reinforcing dosage dependence.

Supporting detail by category: - **Molecular pathways:** ER-stress / **unfolded protein response (UPR)** — PERK/ATF4–CHOP and GRP94 chaperone axis. Collagen biosynthesis/ secretion pathway (Reactome: Collagen biosynthesis and modifying enzymes, R-HSA-1650814; Collagen trimerization). - **Cellular processes:** Protein misfolding, ER retention, UPR activation, **apoptosis** of growth-plate chondrocytes; impaired chondrocyte secretory function. - **Protein**

dysfunction: Dominant-negative misassembly of type II procollagen trimers; intracellular aggregation/retention. - **Metabolic/biochemical:** Reduced secreted type II collagen and aggrecan; defective cartilage matrix composition. - **Immune involvement:** None (non-immune disorder). - **Tissue damage:** ER-stress-driven apoptotic loss of chondrocytes in growth plate. - **GO terms:** GO:0030199 (collagen fibril organization); GO:0006986 (response to unfolded protein); GO:0034976 (response to ER stress); GO:0036503 (ERAD pathway); GO:0006915 (apoptotic process); GO:0001958 (endochondral ossification). - **CL terms:** CL:0000138 (chondrocyte); CL:0000057 (fibroblast, for cDNA studies).

7. Anatomical Structures Affected

- **Organ/system level: Skeletal system** is primarily affected — the **axial skeleton (vertebrae/spine)** and **appendicular skeleton (long bones, hands/feet)**. Secondary: **respiratory system** (hypoplastic lungs due to small thorax).
- **Tissue level: Cartilage / connective tissue** — specifically the **cartilage growth plate** and developing **vertebral bodies**; nucleus pulposus and, in milder/viable type II collagenopathies, the vitreous body are type II collagen–containing tissues.
- **Cell level: Growth-plate chondrocytes** (CL:0000138) are the primary affected cell population.
- **Subcellular level: Rough endoplasmic reticulum** (GO:0005791; GO:0005788 ER lumen) is the key compartment (distended, retains misfolded procollagen); the secretory pathway broadly.
- **UBERON terms:** UBERON:0002481 (bone tissue); UBERON:0001474 (bone element); UBERON:0001075 (vertebral column/axial skeleton); UBERON:0002515 (growth plate / cartilage growth plate); UBERON:0001091 (cartilage); UBERON:0002228 (rib cage/thorax); UBERON:0002048 (lung).
- **Lateralization: Bilateral and symmetric** (generalized skeletal dysplasia).

8. Temporal Development

- **Onset: Congenital / prenatal**; typically detected on second-trimester prenatal ultrasound (short, bowed long bones; narrow thorax; platyspondyly) or at birth.
- **Onset pattern:** Congenital, non-acute (developmental).
- **Progression:** The lethal form is **non-progressive in the usual sense** because death occurs perinatally. In rare survivors and the milder SPD end, disproportionate short stature and

skeletal/spinal complications persist lifelong; early-onset spondyloarthropathy/osteoarthritis is seen in milder allelic conditions.

- **Disease course:** Self-limited by perinatal lethality in classic PLSD-T; chronic lifelong in survivors/SPD.
- **Critical period:** The vulnerable window is fetal/early neonatal (endochondral ossification and lung development). No intervention window currently alters the course.

9. Inheritance and Population

- **Inheritance: Autosomal dominant**; the vast majority are **de novo** heterozygous mutations. Apparent recessive-like sibling recurrence is explained by **parental germline/somatic mosaicism** (Finding 3).
- **Penetrance: Complete** for the constitutional heterozygous (non-mosaic) state.
- **Expressivity:** Variable along the **PLSD-T ↔ SPD spectrum**, driven largely by mutation identity and mosaic dosage.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Mosaicism:** Documented germline and somatic mosaicism (P 22711552). P 22495950
- **Founder effects / consanguinity:** None established; consanguinity is not relevant to a dominant-negative de novo disorder (note: rare *recessive* COL2A1 disorders exist but are distinct from PLSD-T).
- **Carrier frequency:** Not applicable (dominant, de novo).
- **Epidemiology: Very rare** — exact prevalence/incidence not established; reported as isolated cases and small series worldwide (Zankl et al. assembled ~10 cases). No ethnic predilection; essentially sporadic global distribution.
- **Sex ratio:** No strong sex bias reported (autosomal).

10. Diagnostics

- **Imaging (primary diagnostic modality): Radiographs** show severe **platyspondyly**, **micromelia**, **brachydactyly**, and characteristic **metaphyseal changes**; **prenatal ultrasound** detects short/bowed long bones and narrow thorax. The radiographic pattern (metaphyseal changes + brachydactyly) distinguishes PLSD-T from other platyspondylic

lethal dysplasias (e.g., thanatophoric dysplasia) and from helical-domain type II collagenopathies.

- **Genetic testing (confirmatory): Single-gene *COL2A1* sequencing** (targeting the C-propeptide-encoding exons) or **skeletal dysplasia gene panels**; **WES/WGS** are effective and increasingly first-line for undiagnosed lethal dysplasias. **Chromosomal microarray/ karyotype/FISH are not informative** (no copy-number/structural lesion). Test for **parental mosaicism** when recurrence-risk counseling is needed.
- **Histopathology:** Growth-plate examination shows disorganized chondrocytes with **distended rough ER** and abnormal matrix (as modeled in mice).
- **Biomarkers / labs:** No specific blood/urine biomarker; diagnosis is radiographic + molecular.
- **Clinical criteria / differential diagnosis:** Differentials include **thanatophoric dysplasia** (FGFR3), **achondrogenesis type II / hypochondrogenesis** (helical-domain COL2A1), **fibrochondrogenesis**, and **spondyloperipheral dysplasia** (allelic, milder). C-propeptide localization of the mutation is the molecular distinguisher.
- **Screening:** No population newborn screening. **Prenatal ultrasound** and, in families with a known mutation or documented parental mosaicism, **targeted prenatal molecular testing / preimplantation genetic testing** apply.

11. Outcome/Prognosis

- **Survival/mortality:** Generally **perinatally lethal** (stillbirth or neonatal death from pulmonary hypoplasia/respiratory failure); **a few long-term survivors reported** (P 15643621).
- **Life expectancy:** Days or less for classic cases; indefinite but with severe disability for rare survivors / SPD.
- **Morbidity (survivors/SPD):** Profound disproportionate short stature, spinal deformity, restrictive respiratory disease, and orthopedic complications; early spondyloarthropathy at the mild allelic end.
- **Complications:** Respiratory failure (primary), spinal cord/foramen magnum compression risk, feeding difficulties.
- **Prognostic factors: Degree of thoracic hypoplasia** and **mutation dosage/mosaic status** are the main determinants; mosaicism predicts the milder (viable) SPD outcome.

12. Treatment

- **No disease-modifying therapy exists.** Management is **supportive/palliative**.
- **Perinatal/neonatal care:** Palliative or intensive respiratory support depending on severity and family wishes; most cases are not survivable.
- **Survivors/SPD:** Multidisciplinary skeletal-dysplasia care — respiratory support, orthopedic/spine management, hearing/vision surveillance (type II collagen tissues), physical/occupational therapy, pain management.
- **Advanced/experimental therapeutics:** None approved. Conceptually, the ER-stress mechanism (Findings 2, 4) suggests **UPR-modulating or allele-specific/ASO knockdown strategies** as theoretical future targets, but none are in clinical use.
- **Pharmacogenomics / targeted / cell / gene therapy / immunotherapy:** Not applicable at present.
- **NCIT suggestions:** NCIT:C15277 (Supportive Care); NCIT:C15431 (Palliative Care); NCIT:C15329 (Physical Therapy); NCIT:C15215 (Respiratory Therapy).

13. Prevention

- **Primary prevention:** Not possible for de novo mutations. Where parental mosaicism is known, **reproductive options** (PGT, prenatal diagnosis, gamete donation) reduce recurrence.
- **Secondary prevention / early detection: Prenatal ultrasound** screening; targeted prenatal molecular testing in at-risk pregnancies.
- **Genetic counseling (central):** Counsel that most cases are de novo with low recurrence, but **germline/somatic mosaicism confers a non-negligible recurrence risk** and can manifest as mild parental SPD
  (
  P 22711552
  P 22495950
).
  P 21356074
  Counseling also addresses pregnancy-associated and newborn risks when a mosaic parent is affected.
- **Immunization / public health / environmental measures:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *COL2A1* is highly conserved. Mouse *Col2a1* (NCBI Gene 12824; *Mus musculus*, NCBI:txid10090) is the key ortholog used for modeling.
- **Natural animal disease:** The mouse **disproportionate micromelia (*Dmm*)** allele is a naturally arising *Col2a1* C-propeptide mutation causing a dysplasia analogous to human C-propeptide collagenopathy
(P 18678883).
No well-characterized naturally occurring companion-animal homolog of PLSD-T specifically is established in OMIA.
- **Comparative biology / conservation:** The ER-stress/misassembly mechanism is evolutionarily conserved — mouse C-propeptide mutants reproduce the human disease and its UPR/apoptosis mechanism, confirming conserved pathophysiology.

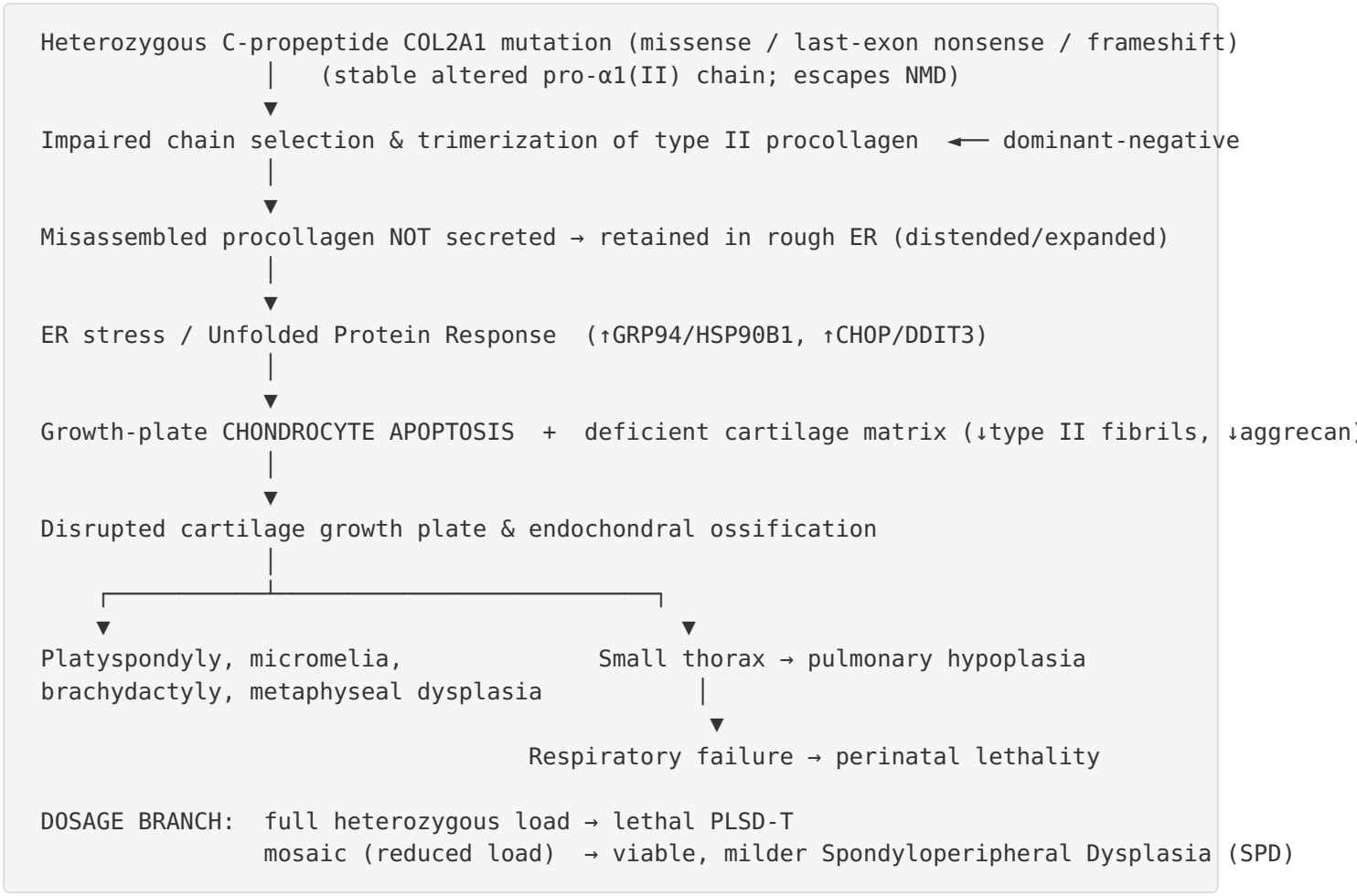
15. Model Organisms

Model	Allele / type	Zygosity → phenotype	Key features	PMID
Mouse (ENU)	*Col2a1* **p.D1469A** (missense, C-propeptide)	Semidominant; het = mild, hom = lethal	Disrupted secretion, distended rough ER, ↑*Grp94*/*Chop*	21538020
Mouse (ENU)	*Col2a1* **p.Y1391S** (missense, C-propeptide)	Semidominant; hom = lethal dysplasia	ER expansion, ER-stress genes up, **severe growth-plate apoptosis**	26545783
Mouse (classic)	**disproportionate micromelia (*Dmm*)** C-propeptide allele	Semidominant; het = dwarfism, hom = lethal	Impaired procollagen trimer assembly/ secretion; thin, sparse fibrils; ↓aggrecan	18678883

- **Model type:** Mammalian (mouse); genetic (ENU-induced and spontaneous point mutations); humanized-residue correspondence (D1469A, Y1391S match human PLSD-T residues).
- **Phenotype recapitulation:** Excellent — homozygotes reproduce lethal micromelia, spondylar/pelvic dysplasia, brachydactyly, and the ER-stress/apoptosis mechanism; heterozygotes model milder/dosage-reduced disease (parallel to human mosaic SPD).

- **Limitations:** Mouse homozygosity is required for the full lethal phenotype (humans are heterozygous, dominant-negative), reflecting species differences in dosage sensitivity; models capture mechanism well but are not perfect genotype matches.
- **Resources:** MGI (*Col2a1*, *Dmm* allele); IMPC; published ENU lines.

Mechanistic Model / Interpretation



The model is notable for being **mechanistically closed** from genotype to phenotype and corroborated in three independent animal models. The central, actionable insight is that PLSD-T is an **ER-storage/UPR-driven apoptotic chondrodysplasia**, placing it mechanistically alongside other collagenopathies and metaphyseal dysplasias where misfolded matrix protein triggers chondrocyte ER stress — and distinguishing it from haploinsufficiency-based skeletal disorders.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
15643621	*Dominant negative C-propeptide COL2A1 mutations cause PLSD-T; novel type 2 collagenopathy subfamily*	Human genetic (case series)	Findings 1, 5 (genotype, phenotype definition)
26545783	*ER stress-mediated apoptosis in a Col2a1 mutant mouse resembling PLSD-T*	Mouse model	Findings 2, 4 (ER stress, apoptosis)
21538020	*ENU missense in Col2a1 C-propeptide creates a mouse model of PLSD-T*	Mouse model	Findings 2, 4 (ER retention, UPR, semidominance)
18678883	*Heterozygous disproportionate micromelia (dmm) mouse*	Mouse model	Finding 4 (procollagen assembly/secretion defect, dosage)
22711552	*PLSD-T in two sibs from normal parents (germline mosaicism)*	Human genetic	Finding 3 (germline mosaicism)
22495950	*SPD as mosaic form of PLSD-T in mother and fetus*	Human genetic	Finding 3 (somatic mosaicism; PLSD-T↔SPD spectrum)
17726487	*Czech dysplasia; Y1391C in SPD*	Human genetic	Supports PLSD-T↔SPD continuum (Y1391C in both)
21356074	*Prenatal management of SPD with C-propeptide mutation*	Human clinical	Genotype–phenotype (SPD–Torrance spectrum); counseling
8723097	*COL2A1 mutation presenting as spondyloperipheral dysplasia*	Human genetic	C-terminal (C-propeptide) frameshift → SPD spectrum
8157027	*The type II collagenopathies: a spectrum*	Review	Places PLSD-T in the type II collagenopathy family
31755234, 32896647	*Biallelic/recessive COL2A1 disorders*	Human genetic	Contrast: distinct recessive type II collagenopathies (NOT PLSD-T)

All findings are convergent; no paper in the reviewed set contradicts the core mechanism. The recessive-COL2A1 papers are included to **delimit** PLSD-T (dominant, C-propeptide, dominant-negative) from mechanistically distinct recessive type II collagenopathies.

Limitations and Knowledge Gaps

1. **Epidemiology is unquantified.** No reliable prevalence/incidence figures exist; the literature is limited to isolated cases and a ~10-case series. Sex ratio and any population differences are undetermined.
2. **Small human N.** Genotype–phenotype correlations (including the PLSD-T↔SPD boundary) rest on a handful of patients; the precise determinants of lethal vs. viable outcome beyond mosaic dosage are not fully resolved.
3. **Mechanism confirmed mainly in mouse.** The ER-stress/apoptosis causal chain is directly shown in mouse models; human chondrocyte in-vivo confirmation is limited to histopathology, and the final step (thorax → pulmonary hypoplasia → death) is inferred rather than experimentally dissected.
4. **No molecular profiling (omics) data** specific to human PLSD-T chondrocytes (transcriptomics/proteomics/single-cell) were identified.
5. **No therapeutics.** UPR-modulating or allele-specific strategies are only conceptual; no preclinical efficacy data exist for PLSD-T.
6. **Modifier genes and epigenetics** are essentially unexplored.

Proposed Follow-up Experiments / Actions

1. **Single-cell/spatial transcriptomics of growth plates** from the D1469A/Y1391S mouse models to map the UPR→apoptosis trajectory at cell-type resolution (GEO deposition).
2. **Preclinical UPR-modulation trials** (e.g., chemical chaperones such as 4-PBA/TUDCA, or genetic CHOP/DDIT3 knockdown) in the established mouse models to test whether dampening ER stress rescues growth-plate apoptosis and skeletal phenotype.
3. **Allele-selective silencing (ASO/siRNA)** proof-of-concept targeting the mutant *COL2A1* transcript in patient-derived or knock-in chondrocytes/iPSC-chondroid organoids.
4. **International PLSD-T/SPD registry** to quantify prevalence, natural history of survivors, mosaicism-associated recurrence risk, and genotype–phenotype boundaries.

5. **Deep phenotyping of parental mosaicism** (sensitive sequencing of multiple parental tissues/gametes) to refine recurrence-risk counseling.
6. **Structural/biophysical study** (UniProt P02458 C-propeptide; AlphaFold + experimental PDB) of how specific residue substitutions impair trimer nucleation, to build a predictive severity model.

Report compiled from 5 iterations of autonomous investigation; 5 confirmed findings; 13 papers reviewed. Causal gene named in first paragraph: COL2A1.
